Assignment 2: Pathfinding and MCTS

Weight: 15pts

Due date: December 14, 2025 11:59PM AEST

Assigment Type: Individual

Assignment Details:

This assignment has 2 parts. The first part is to implement the A* Algorithm so whenever you clicked a target position, the frog (same frog from assignment 1) can move to that target position (with a steering behavior) by using the shortest path possible. For the second part of the assignment, you sould implement the MCTS algorithm to play the Connect4 game.

Part 1: A* Pathfinding

For Part 1, you should begin by studying the Pathfinding game object to understand how it works. Note that the frog doesn’t actually use pathfinding yet from your assignment 1 -- it's still configured to move directly towards the right-clicked location via Arrive.

<u>Make the frog follow the A* path</u>

When the player right-clicks a location, the frog should use the shortest path to move to the corresponding target location. The shortest path should have a green color every time you clicked to a target location so that the frog follows the A* path, rather than moving directly towards the clicked position. Remember: the frog needs to follow the sortest path to arrive at the target location clicked.

Notes:

- You’ll need to think about when the frog should move on to the next waypoint in the path.
- You should aim to make the path following look as natural as The frog should still move via steering behaviours, and you think carefully about which steering behaviours to use. The frog shouldn’t collide with corners very often.

<u>Update the A* algorithm to allow diagonal neighbours</u>

There is an option to consider only cardinal movements (up, down, left, right) when determining the neighbours of a node. But this yields very clunky-looking paths; see Figure 2 (left) for an illustration. You need also to treat diagonal nodes as neighbours too, which should yield more natural-looking paths.

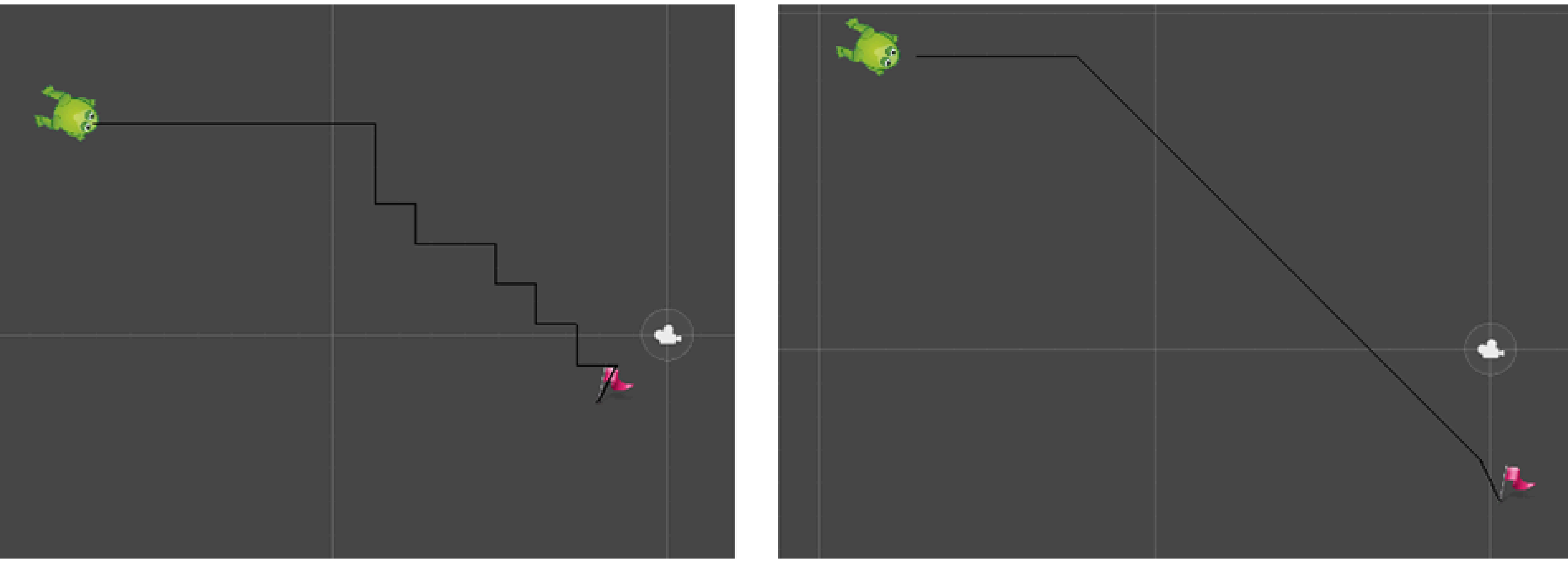
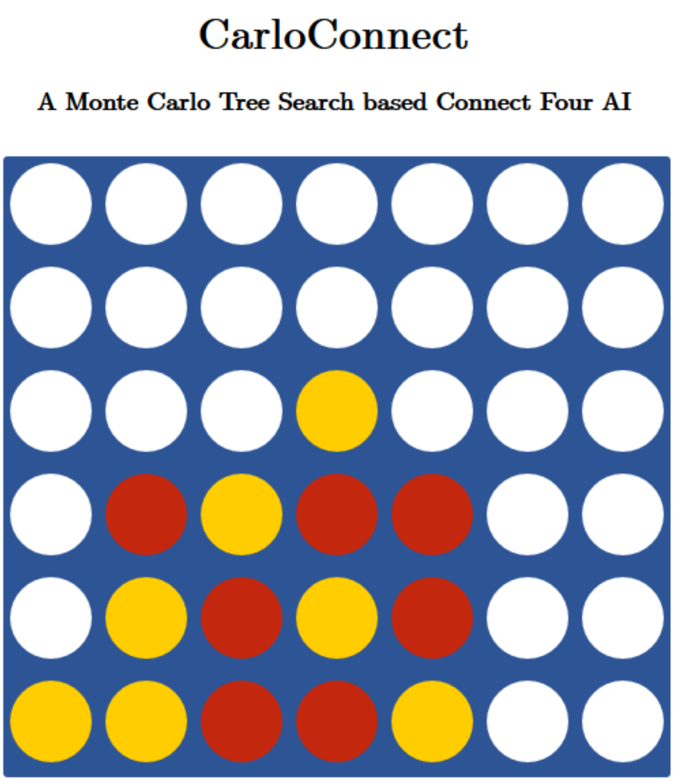


Figure 2: Examples of the types of paths found with diagonal neighbours disabled (left) and enabled (right).

Part 2: MCTS for Connect4

For this part of the assignment, you will implement the *Connect4 (see image below)* using MCTS. [You are given this starter code](#) where the game behavior is 2 human players playing against each other; your task is to update the game so instead of 2 players, your task will be:

- Implement a scene for human to play against an AI agent.
- Implement a scene for 2 AI agents to play against each other.



Marking Guide

Note: The below guide is only approximate. The teaching staff may also deduct points for not meeting implied, common sense requirements, e.g., you should not modify the main pathfinding scene to hide issues with the pathfinding, and you try strive to write efficient code that follows good programming practices.

Part 1: A* Pathfinding

Make the frog follow the A* path (4)

- 4 marks for behaviour matching the standard description.
- 2 marks if the frog roughly follows the path but there are one or more issues.
- 1 mark if path following is badly broken but there is evidence of a genuine attempt at the problem.
- 0 marks if not attempted or very minimal attempt.

Implement a basic Monte Carlo agent (9)

- 8-9 marks for a perfect implementation.
- 5-7 marks if the implementation is mostly correct and the agent plays much better than random, but there is a small issue, e.g., the agent neglects one of the available moves, the simulation budget isn't adhered to, etc.
- 4-2 marks (at the marker's discretion) if the agent is far weaker than expected due to a significant bug.
- 0 marks if not attempted or very minimal attempt.

Creativity (2pts)

- Design and color choice
- AI Agents smart features added
- Any other interesting non mentionned idea

Criteria	Ratings		Pts
A* Pathfinding	**4 pts** **Full Marks** Make the frog follow the A* path correctly. 2-3 marks if the frog roughly follows the path but there are one or more issues, 1 mark if path following is badly broken but there is evidence of a genuine attempt at the problem. 0 marks if not attempted or very minimal attempt.	**0 pts** **No Marks** not attempted or very minimal attempt.	4 pts
MCTS for Connect4	**9 pts** **Full Marks** Implement a basic Monte Carlo agent (4 marks) 4 marks for a perfect implementation. 3 marks if the implementation is mostly correct 1-2 marks (at the marker's discretion) if the agent is far weaker than expected due to a significant bug. 0 marks if not attempted or very minimal attempt.	**0 pts** **No Marks** not attempted or very minimal attempt.	9 pts
Creativity	**2 pts** **Full Marks** Design and color choice AI Agents smart features added Any other interesting non mentionned idea	**0 pts** **No Marks** not attempted or very minimal attempt.	2 pts
			Total Points: 15